Example: Complete QP Solution

Original Function:

Minimize 0.5 x^2 + 3x + 4y

Matrix Form: Minimize ½ [x^T P x] + q^T x

```
P = [ 1  0 ]
    [ 0  0 ]

q = [ 3 ]
    [ 4 ]
```

Constraints Gx ≤ h:

x ≥ 0 => -x ≤ 0

y ≥ 0 => -y ≤ 0

x + 3y ≥ 15 => -x - 3y ≤ -15

2x + 5y ≤ 100

3x + 4y ≤ 80

G =

```
[-1  0 ]
[ 0 -1 ]
[-1 -3 ]
[ 2  5 ]
[ 3  4 ]
```

SOLUTION

Optimal Point: x = 0, y = 5

Constraint Check:

x + 3y = 15 ≥ 15 ✓

2x + 5y = 25 ≤ 100 ✓

3x + 4y = 20 ≤ 80 ✓

Minimum Value:

0.5 x^2 + 3x + 4y = 20

h =

```
[  0 ]
[  0 ]
[-15 ]
[100 ]
[ 80 ]
```